# Area in Polar Coordinates

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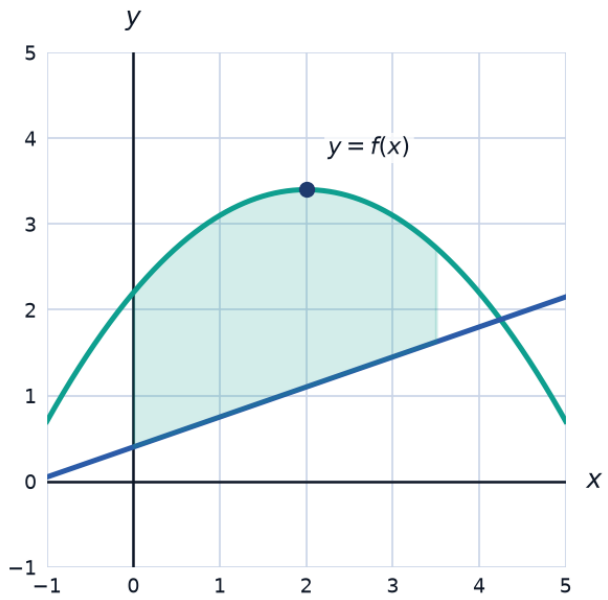
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: The polar area formula**

State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

**2** **[Foundational]** • **Target: The polar area formula**

Find the area enclosed by one full loop of $r = 3\cos\theta$ (i.e. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$).




**3** **[Foundational]** • **Target: Area of a full cardioid**

Find the total area enclosed by the cardioid $r = 1 + \cos\theta$.

**4** **[Foundational]** • **Target: Area between two polar curves**

Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

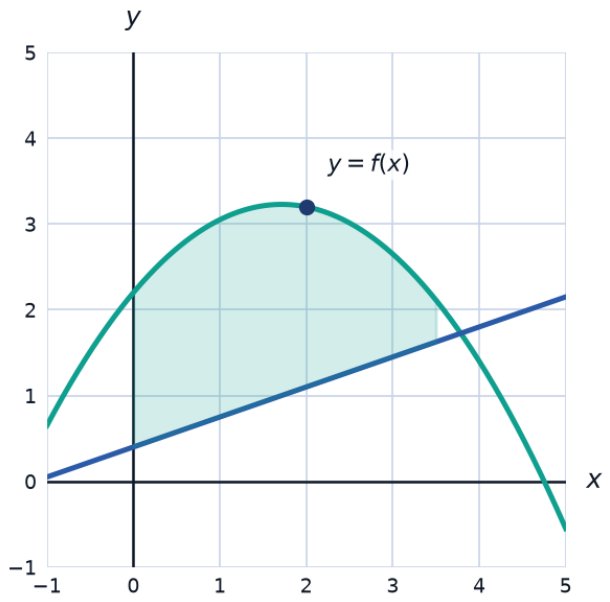
**5** **[Foundational]** • **Target: Area of one petal of a rose curve**

Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in [0, \frac{\pi}{2}]$ where one petal is traced.

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**6** [Foundational] • Target: Setting up area between curves that intersect

Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos\theta$, noting they intersect where $2 = 2 + 2\cos\theta$, i.e. $\theta = \pm\frac{\pi}{2}$.




**7** [Foundational] • Target: Area of a full rose curve

Find the total area enclosed by all $4$ petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Area inside a limaçon

Find the area enclosed by the limaçon $r = 3 + \cos\theta$ (a curve without an inner loop, since the constant exceeds the coefficient).

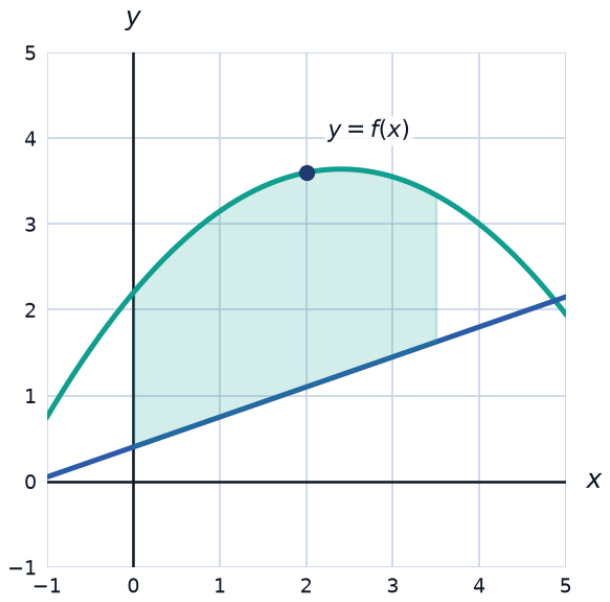
**9** [Intermediate] • Target: Area inside a limaçon

Find the area of the small inner loop of the limaçon $r = 1 + 2\cos\theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right]$.

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**10** [Intermediate] • Target: Setting up area problems (AP-style, no evaluation)

Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in [0, \frac{\pi}{3}]$.




**11** [Intermediate] • Target: Setting up area problems (AP-style, no evaluation)

Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin\theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.

**12** [Intermediate] • Target: Finding intersection points of polar curves

Find all points of intersection of $r = 2\cos\theta$ and $r = 2\sin\theta$ (both circles of radius $1$) by setting the equations equal.

**13** [Intermediate] • Target: Finding intersection points of polar curves

Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.

**14** [Intermediate] • Target: Area using symmetry to simplify computation

Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

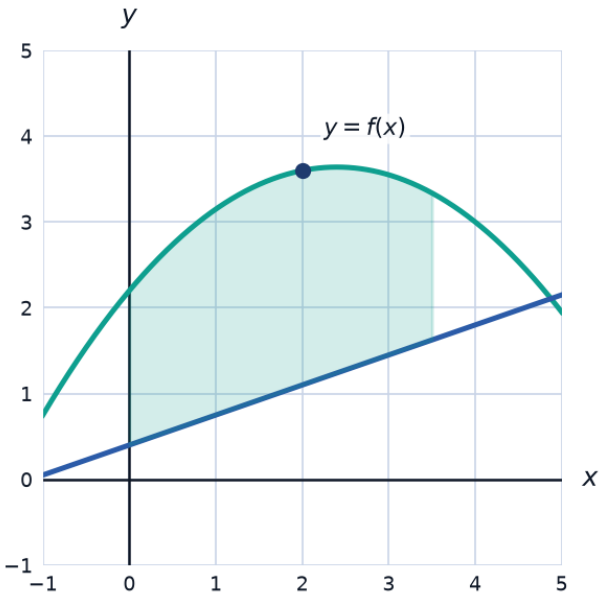
**15** [Challenging] • Target: Area review: full worked example • Cross-Lesson: 1.01 Integration by Parts

Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

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**16** [Challenging] • Target: Area review: full worked example • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Find the area enclosed by $r = 4\sin\theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.




**17** [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Explain why the polar area formula $A = \frac{1}{2}\int r^2\,d\theta$ uses $r^2$ rather than just $r$, by relating it to the area of a thin circular sector.

**18** [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 3.01 Parametric Equations

A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2}\int_0^{2\pi} (2-1)^2\,d\theta$. Explain the error and give the correct setup.

**19** [Challenging] • Target: Area between two polar curves • Cross-Lesson: 3.02 Vector-Valued Functions

Convert the rectangular point $(0, -3)$ to polar coordinates.

**20** [Challenging] • Target: Area of one petal of a rose curve • Cross-Lesson: 4.01 Polar Curves and Derivatives

Convert the polar point $(2, \frac{\pi}{2})$ to rectangular coordinates.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: The polar area formula

Find the area enclosed by $r = 1$ for $\theta \in [0, \frac{\pi}{2}]$.

**a.** $\frac{\pi}{4}$

**b.** $\frac{\pi}{2}$

**c.** $-\frac{7\pi}{4}$

**d.** $\frac{3\pi}{4}$

**22** [Foundational] • Target: Area of a full cardioid

Find the area enclosed by $r = 3$ for $\theta \in [0, \frac{3\pi}{2}]$.

**a.** $\frac{31\pi}{4}$

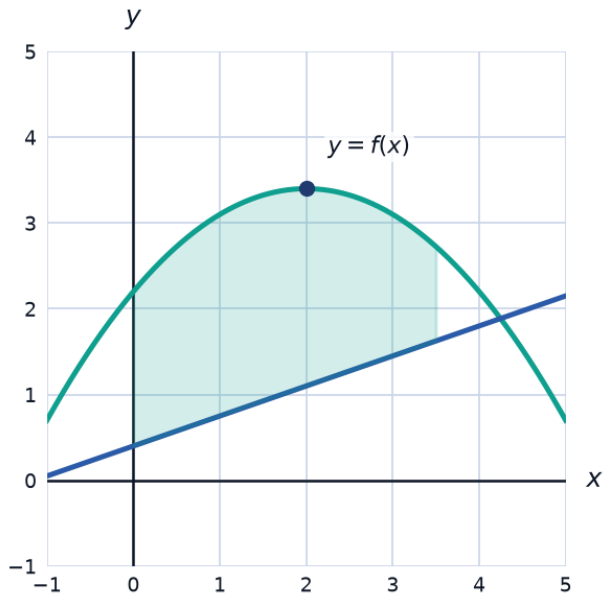
**b.** $\frac{27\pi}{2}$

**c.** $\frac{27\pi}{4}$

**d.** $\frac{29\pi}{4}$

**23** [Foundational] • Target: Area between two polar curves

Find the area enclosed by $r = 1$ for $\theta \in [0, \pi]$.




**a.** $-\frac{3\pi}{2}$

**b.** $\pi$

**c.** $\frac{5\pi}{2}$

**d.** $\frac{\pi}{2}$

**24** [Foundational] • Target: Area of one petal of a rose curve

Find the area enclosed by $r = 6$ for $\theta \in [0, \frac{\pi}{2}]$.

**a.** $18\pi$

**b.** $7\pi$

**c.** $\frac{19\pi}{2}$

**d.** $9\pi$

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**25** [Foundational] • Target: Setting up area between curves that intersect

Find the area enclosed by $r = 9$ for $\theta \in [0, \frac{\pi}{2}]$.

**a.** $\frac{81\pi}{4}$

**b.** $\frac{81\pi}{2}$

**c.** $\frac{89\pi}{4}$

**d.** $\frac{83\pi}{4}$

**26** [Foundational] • Target: Area of a full rose curve

Find the area enclosed by $r = 9$ for $\theta \in [0, \pi]$.

**a.** $81\pi$

**b.** $\frac{83\pi}{2}$

**c.** $\frac{81\pi}{2}$

**d.** $41\pi$

**27** [Foundational] • Target: Area inside a limaçon

Find the area enclosed by $r = 8$ for $\theta \in [0, 2\pi]$.

**a.** $\frac{129\pi}{2}$

**b.** $62\pi$

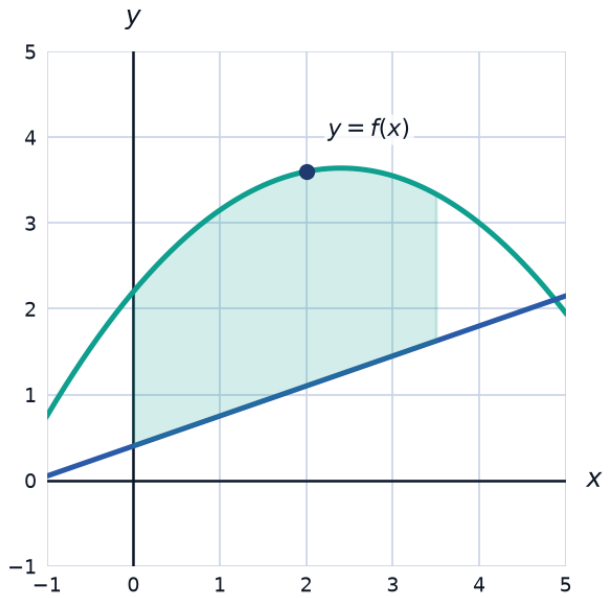
**c.** $64\pi$

**d.** $128\pi$

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**28** [Foundational] • Target: Setting up area problems (AP-style, no evaluation)

Find the area inside the circle $r = 9$ but outside the circle $r = 4$.




**a.** $-7\pi$

**b.** $25\pi$

**c.** $77\pi$

**d.** $65\pi$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Finding intersection points of polar curves

Find the area inside the circle $r = 8$ but outside the circle $r = 5$.

**a.** $39\pi$

**b.** $59\pi$

**c.** $9\pi$

**d.** $-17\pi$

**30** [Intermediate] • Target: Area using symmetry to simplify computation

Find the area inside the circle $r = 13$ but outside the circle $r = 6$.

**a.** $163\pi$

**b.** $-23\pi$

**c.** $49\pi$

**d.** $133\pi$

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**31** [Intermediate] • Target: Area review: full worked example

Find the area inside the circle $r = 10$ but outside the circle $r = 5$.

**a.** $-15\pi$

**b.** $75\pi$

**c.** $95\pi$

**d.** $25\pi$

**32** [Intermediate] • Target: Comparing polar area to rectangular area (conceptual)

Find the area inside the circle $r = 8$ but outside the circle $r = 3$.

**a.** $55\pi$

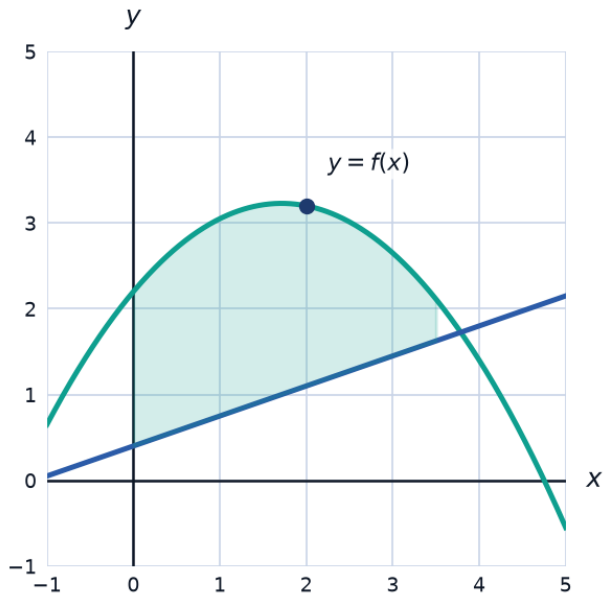
**b.** $61\pi$

**c.** $25\pi$

**d.** $-\pi$

**33** [Intermediate] • Target: Sector Area Formula & Single Curves

Find the area inside the circle $r = 10$ but outside the circle $r = 3$.




**a.** $49\pi$

**b.** $97\pi$

**c.** $91\pi$

**d.** $\pi$

**34** [Intermediate] • Target: Area Between Two Polar Curves

Find the total area enclosed by the curve $r = 8 + 7\sin\theta$.

**a.** $113\pi$

**b.** $\frac{177\pi}{2}$

**c.** $\frac{179\pi}{2}$

**d.** $\frac{225\pi}{2}$

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**35** [Intermediate] • Target: Common Interior of Two Curves

Find the total area enclosed by the curve $r = 5 + 9\sin\theta$.

**a.** $98\pi$

**b.** $106\pi$

**c.** $\frac{133\pi}{2}$

**d.** $\frac{131\pi}{2}$

**36** [Intermediate] • Target: Integration Limits & Full-Period Curves

Find the total area enclosed by the curve $r = 3 + 5\cos\theta$.

**a.** $34\pi$

**b.** $32\pi$

**c.** $\frac{45\pi}{2}$

**d.** $\frac{43\pi}{2}$

**37** [Intermediate] • Target: The polar area formula

Find the area of one petal of $r = 4\sin(7\theta)$.

**a.** $\frac{23\pi}{28}$

**b.** $\frac{8\pi}{7}$

**c.** $\frac{16\pi}{7}$

**d.** $\frac{4\pi}{7}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Area of a full cardioid • Cross-Lesson: 3.01 Parametric Equations

Find the area of one petal of $r = 4\sin(2\theta)$.

**a.** $\frac{5\pi}{2}$

**b.** $8\pi$

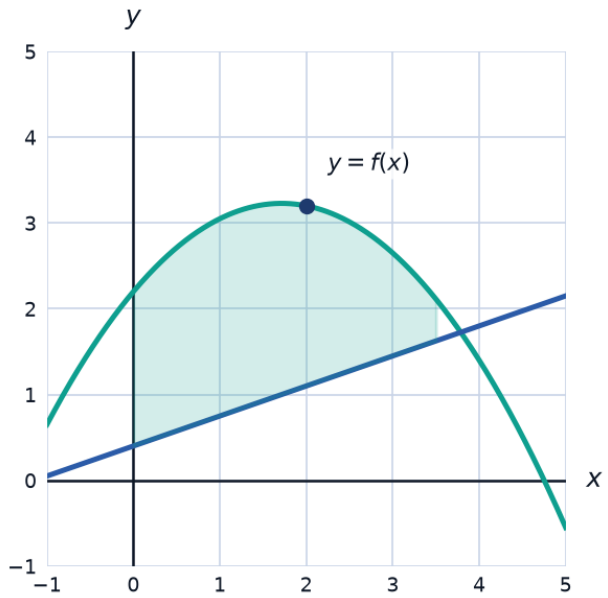
**c.** $4\pi$

**d.** $2\pi$

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**39** [Challenging] • Target: Area between two polar curves • Cross-Lesson: 3.02 Vector-Valued Functions

Find the area inside the circle $r = 11$ but outside the circle $r = 7$, for $\theta \in [0, \pi]$ only (half the full region).




**a.** $8\pi$

**b.** $72\pi$

**c.** $36\pi$

**d.** $37\pi$

**40** [Challenging] • Target: Area of one petal of a rose curve • Cross-Lesson: 4.01 Polar Curves and Derivatives

Find the area inside the circle $r = 3$ but outside the circle $r = 1$, for $\theta \in [0, \pi]$ only (half the full region).

**a.** $5\pi$

**b.** $8\pi$

**c.** $4\pi$

**d.** $2\pi$

**41** [Challenging] • Target: Sector Area Formula & Single Curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Which integral formula calculates the area of a region bounded by the polar curve $r = f(\theta)$ from $\alpha$ to $\beta$?

**a.** $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$

**b.** $\int_{\alpha}^{\beta} r^2 d\theta$

**c.** $\frac{1}{2}\int_{\alpha}^{\beta} r d\theta$

**d.** $\pi\int_{\alpha}^{\beta} r^2 d\theta$

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**42** [Challenging] • Target: Area Between Two Polar Curves • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Which integral represents the area inside $r = R(\theta)$ and outside $r = r(\theta)$?

**a.** $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta)^2 - r(\theta)^2)d\theta$

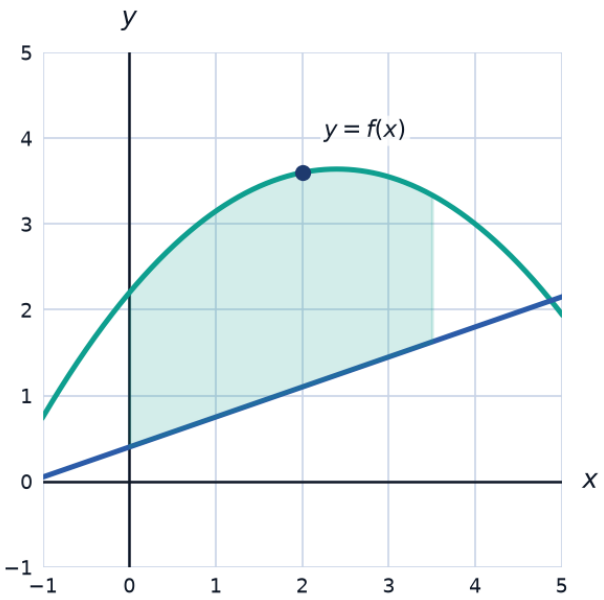
**b.** $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta) - r(\theta))^2 d\theta$

**c.** $\int_{\alpha}^{\beta}(R^2 - r^2)d\theta$

**d.** $\frac{1}{2}\int_{\alpha}^{\beta}R(\theta)r(\theta)d\theta$

**43** [Challenging] • Target: Area Between Two Polar Curves • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Set up the integral for the area inside $r = 6\cos\theta$ and outside $r = 3$.




**a.** $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(36\cos^2\theta - 9)d\theta$

**b.** $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(6\cos\theta - 3)^2 d\theta$

**c.** $\int_{0}^{\pi/3}(36\cos^2\theta - 9)d\theta$

**d.** $\frac{1}{2}\int_{0}^{\pi/3}(6\cos\theta - 3)d\theta$

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**44** [Challenging] • Target: Common Interior of Two Curves • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the area inside both $r = 2$ and $r = 4\sin\theta$ in the first quadrant.

**a.** $\frac{4\pi}{3} - \sqrt{3}$

**b.** $2\pi$

**c.** $\frac{2\pi}{3}$

**d.** $4\sqrt{3}$

**45** [Challenging] • Target: Integration Limits & Full-Period Curves • Cross-Lesson: 1.01 Integration by Parts

What is the range of $\theta$ required to trace the limaçon $r = 2 + \cos\theta$ exactly once?

**a.** $[0, 2\pi]$

**b.** $[0, \pi]$

**c.** $[0, 4\pi]$

**d.** $[-\frac{\pi}{2}, \frac{\pi}{2}]$

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